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PROJECT:

CMMS DEVELOPMENT PROJECT

PC58963	OAPRMMPF999061	A 1

DOCUMENT DESCRIPTION:

ASSET INTEGRITY

GUIDELINE TO CREATE OFFLINE PREDICTIVE MAINTENANCE WORK ORDER PROCEDURE

Status / Revision	Date (DD-MMM-YYYY)	No. of Pages	Written by	Checked by	Approved by	Approval for issue
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INFORMATION ON STATUS:

P	**Preliminary for Information**
I	**Inter Discipline Checking**
C	**For Comments and Approval**
V	**Valid for Construction**
A	**Approved for Construction**
X	**"As Built"**
EPM	**Engineering Project Manager or Assigned Substitute**

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REVISION STATUS / SUMMARY OF CHANGES

REVISION	REVISED CHAPTERS	REVISION DESCRIPTION	REASON FOR REVISION

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1. INTRODUCTION

Predictive maintenance, or CBM (Condition-Based Maintenance), is a strategy focused on monitoring the operational condition of assets with the aim of increasing reliability, optimizing resources, and preventing unexpected failures. At SBM, WOs (Work Orders) must be created whenever anomalies are identified in offline predictive monitoring reports, such as those from vibration or lubricant analysis, and after a critical analysis made by competent role.

To ensure traceability, standardization, and a reliable maintenance history, it is essential that all CBM-related WOs are correctly and consistently registered in the CMMS (Computerized Maintenance Management System). This guarantees accurate documentation of identified anomalies, technical recommendations, and corrective actions, while also facilitating audits, supporting continuous improvement, and enhancing decision-making based on historical data.

This document provides clear guidelines for creating offline predictive maintenance work orders in the CMMS, detailing the mandatory fields and offering step-by-step instructions to ensure proper data entry. It also defines the responsibilities of the stakeholders involved, ensuring compliance with internal maintenance governance standards.

2. PURPOSE

The purpose of this document is to provide clear guidelines for the creation of predictive maintenance work orders in the CMMS, specifically for offline predictive maintenance activities. It defines the mandatory fields that must be completed, as well as detailed instructions for their proper completion.

3. SCOPE

This document is applicable to:

a) All fleet (Regions 01 and 02)

4. ACRONYMS

CMMS	Computerized Maintenance Management System
CBM	Condition Based Maintenance
WO	Work Order
CM	Corrective Maintenance
BM	Breakdown Maintenance
TAG	Object ID
PCE	Production Critical Equipment
NCE	Non-Critical Equipment
SCE	Safety Critical Equipment
ECE	Environmental Critical Equipment
SECE	Safety and Environment Critical Equipment
RACI	Responsible Accountable Checker and Information

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5. DETAILED INSTRUCTIONS

The effectiveness of the process depends on the correct completion of the mandatory fields in the CMMS. A detailed description of each required field is provided below, along with step-by-step instructions on how to complete them accurately.

5.1 OBJECT NAVIGATOR

The process of creating a WO begins with searching for the equipment in the CMMS for which the WO is to be created.

With the IFS CMMS open on the home screen, search for the equipment TAG (Object ID) for which you want to raise a CM (Corrective Maintenance) WO using the “Equipment Object Navigator” transaction.

As soon as the search window appears on the screen, enter the equipment's TAG (Object ID) and the unit code.

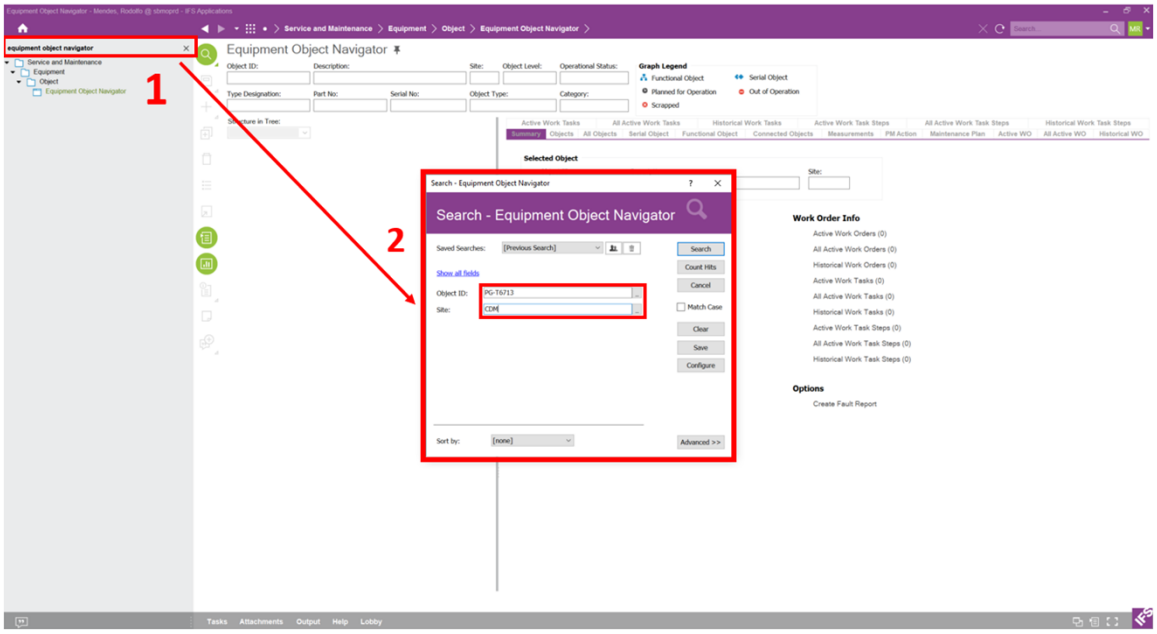


Figure 1 – Equipment Object Navigator

5.2 FUNCTION OBJECT

After performing the search, with the IFS CMMS open on the “Functional Object” screen, right-click to display the “Fault Report” transaction. Then, select this option by clicking the left mouse button.

NOTE 01: A WO can only be raised for equipment registered at the TAG level. The IFS CMMS does not allow the raising of WOs for equipment registered at the Header level.

NOTE 02: Considering ISO 14224 as a reference, all work orders must be created at levels 6 or 8. Where level 6 refers to the package, and level 8 refers to the maintainable component/item. The WO must always be raised for the lowest-level rotating equipment

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where the anomaly was identified.

EXAMPLE 01: In a lubricant analysis where the lubricant services more than one component within the package (Gearbox + Compressor), the WO should be opened at the package level if it is a TAG. If it is a header, prioritize the following order:

A. Driven equipment
B. Driver equipment
C. Transmission equipment

EXAMPLE 02: Vibration analysis detected an issue in the electric motor. The WO should be opened for motors. If it is not available, prioritize the following order:

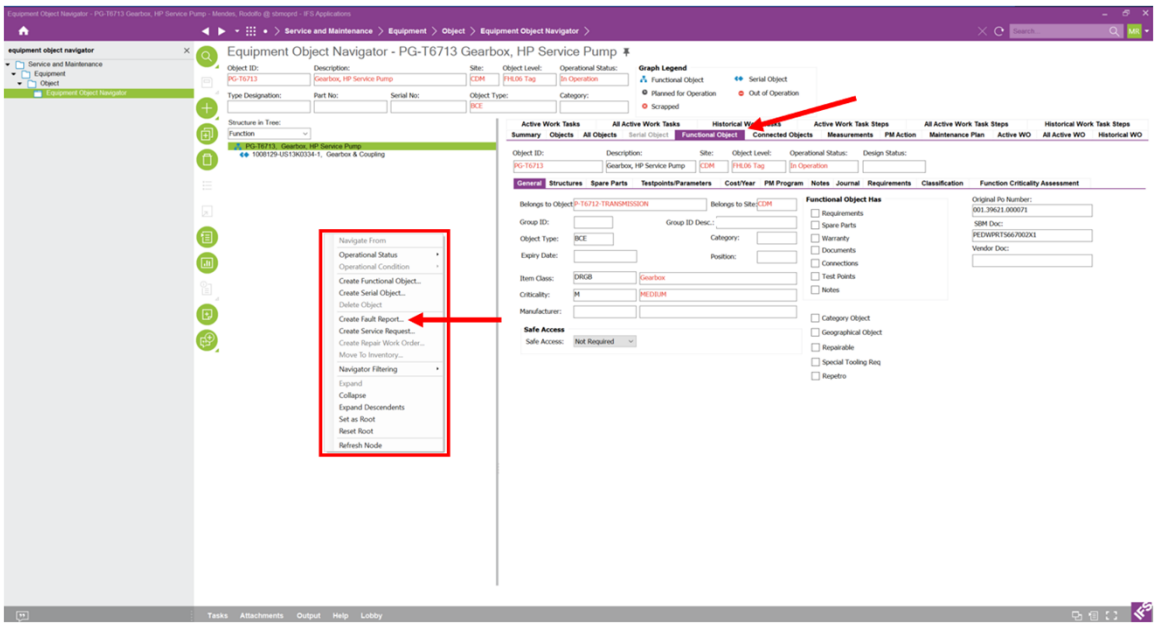


Figure 2 – Equipment Object Navigator Screen

5.3 FALT REPORT FOR OBJECT

On the next screen, all the fields that shall be populated will appear. The fields to be completed are:

- WO Site
- Directive
- Maint. Org.
- Work Type
- External Source
- External Source ID
- Fault Desc.
- Symptom
- Discovery
- Action ID

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- Operational Status

All the fields that shall be populated are highlighted in the image below. A brief description of each field, as well as the instructions for correct completion, can be found afterward.

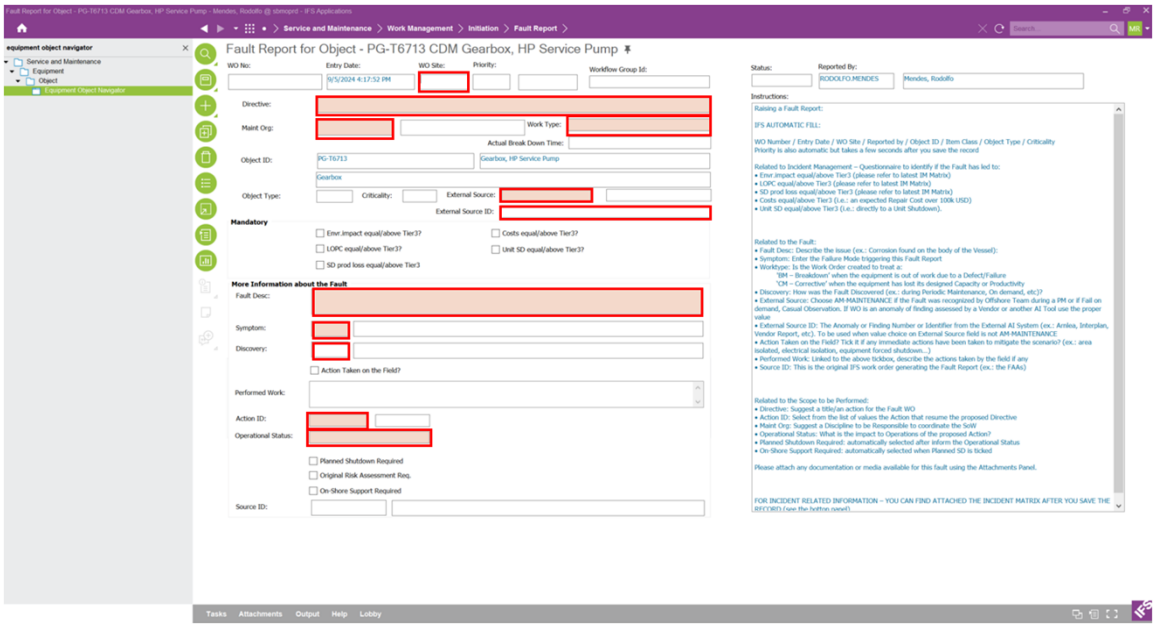


Figure 3 - Mandatory fields to populate the Fault Report

5.4 WO SITE

This field refers to the unit for which the WO is being raised in the CMMS. This field shall be populated with the unit for which the WO will be raised.

CMMS STEP BY STEP

I. Enter the unit code directly into the "WO Site" field, as illustrated in the image below.
II. After that, press “ENTER” to confirm the transaction

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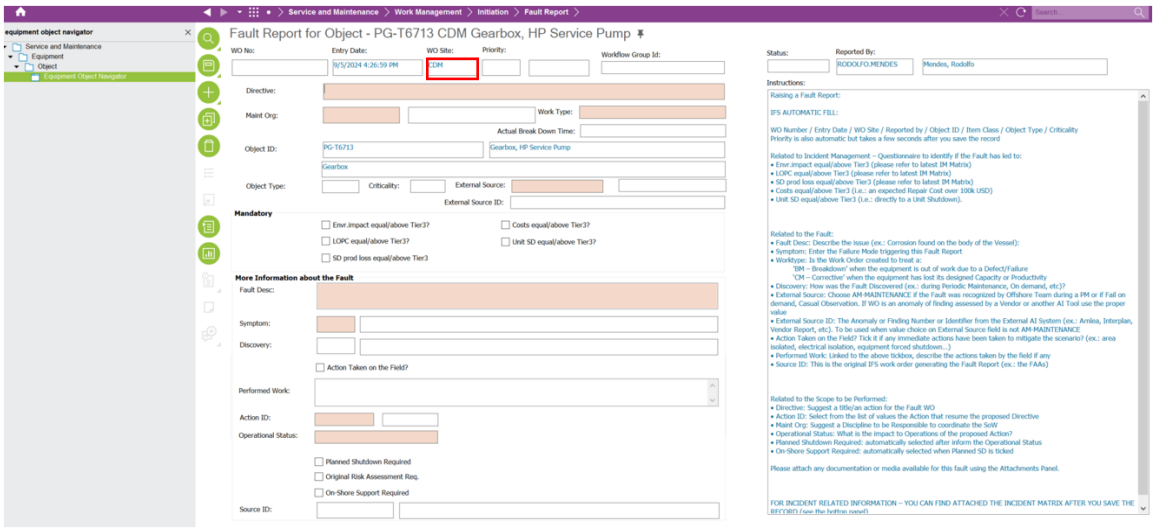


Figure 4 - Field to populate the WO site

5.5 DIRECTIVE

This field refers to the title of the WO. It must be filled in in a standardized manner, using capital letters in the description field and a brief description of the recommended corrective action.

EXAMPLE 01: Anomaly identified through the lubricant analysis monitoring technique (e.g., high particle count) and recommendation for oil replace. Directive: OIL REPLACE.

EXAMPLE 02: Anomaly identified through the vibration analysis monitoring technique (e.g., unbalance) and recommendation for trim balance. Directive: TRIM BALANCE.

CMMS STEP BY STEP

I. Populate the field following the instructions above.

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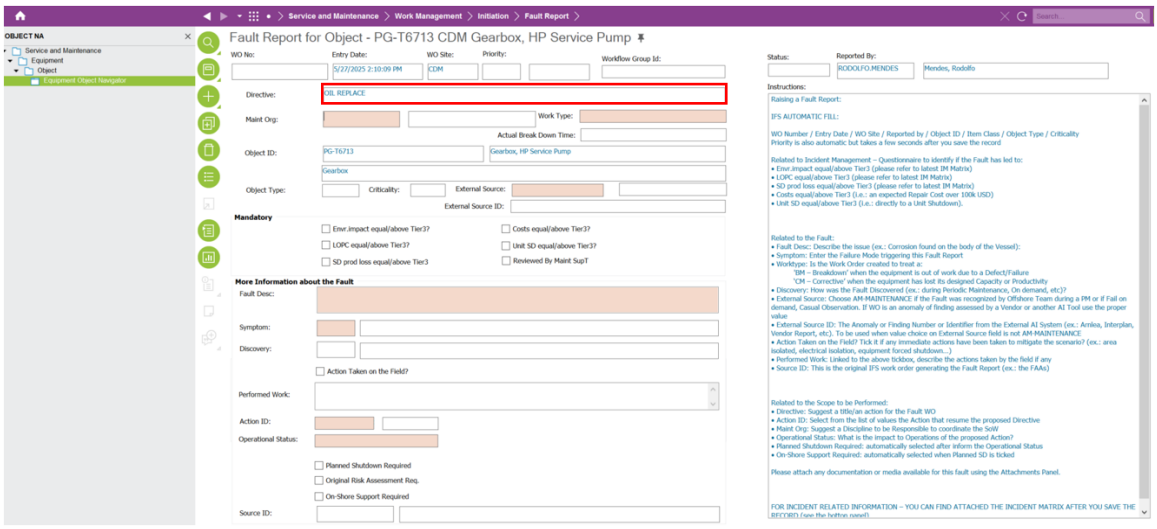


Figure 5 - Field to populate the directive

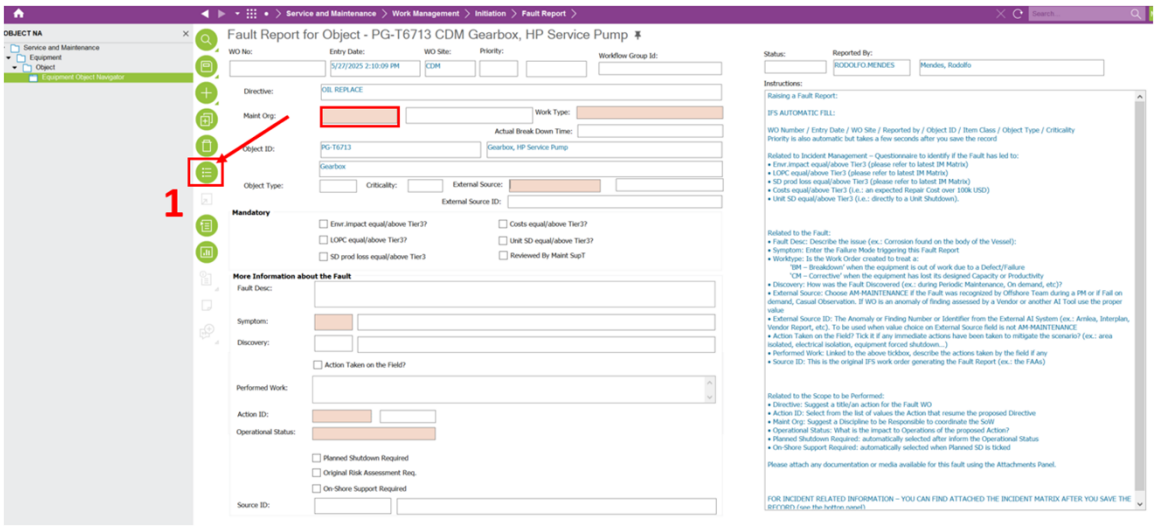
5.6 MAINT. ORG.

This field refers to the discipline responsible for executing the work. To correctly populate this field in the CMMS, the user must consider the type of work as well as the location of the assets in the plant.

For example, WO raised for mechanical equipment located on the Topside should be performed by the “MECHTS” discipline. On the other hand, WO raised for equipment located in the Engine Room should be performed by the “MECHER” discipline.

CMMS STEP BY STEP

I. Click on the "Values List" available in the CMMS to open the selection window.



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Figure 6 - Field to populate the maint. Org

II. Select the desired option from the selection window.

III. After making your selection, click "OK" to confirm the transaction, as shown below

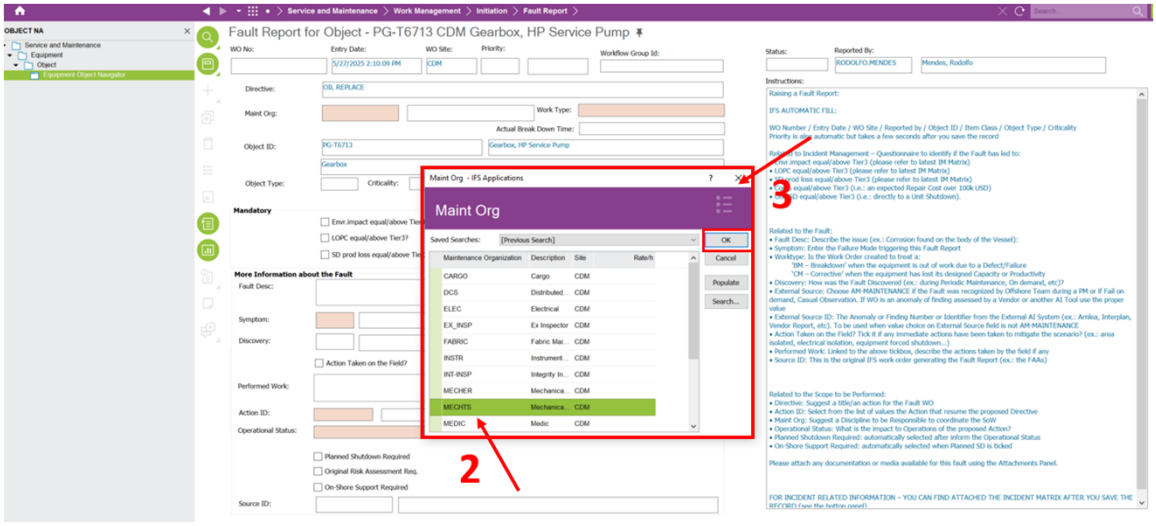


Figure 7 - Confirming the selection of the maint. Org

5.7 WORK TYPE

This field refers to the type of WO that will be raised. There are only two options available for selection within the CMMS: Breakdown Maintenance (BM) and Corrective Maintenance (CM). Work orders originating from predictive analyses should always be raised using the Corrective (CM) option.

CMMS STEP BY STEP

I. Click on the "Values List" available in the CMMS to open the selection window.

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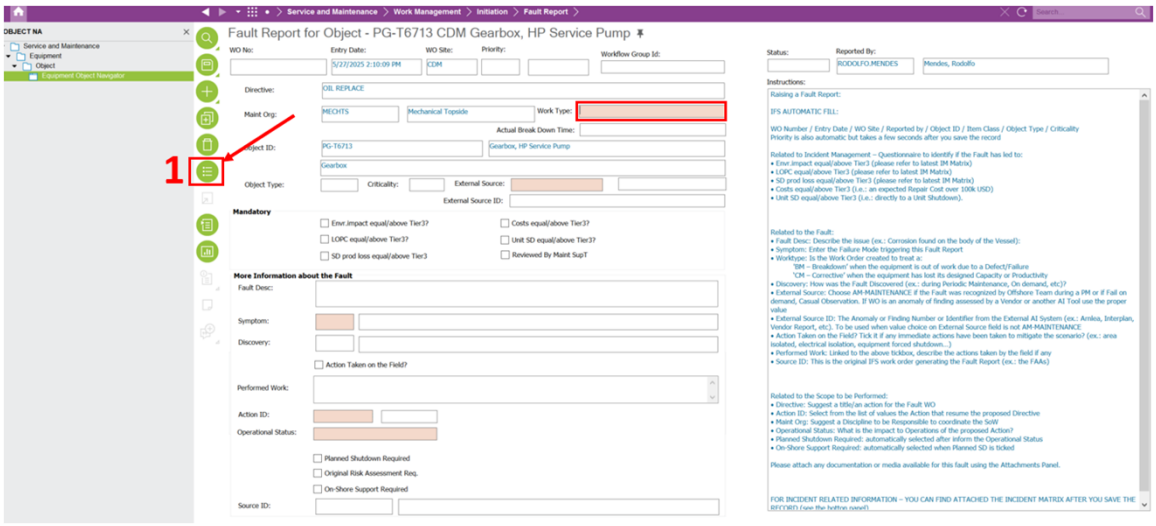


Figure 8 - Field to populate the work type

II. Select the desired option “CM” from the selection window.

III. After making your selection, click "OK" to confirm the transaction, as shown below.

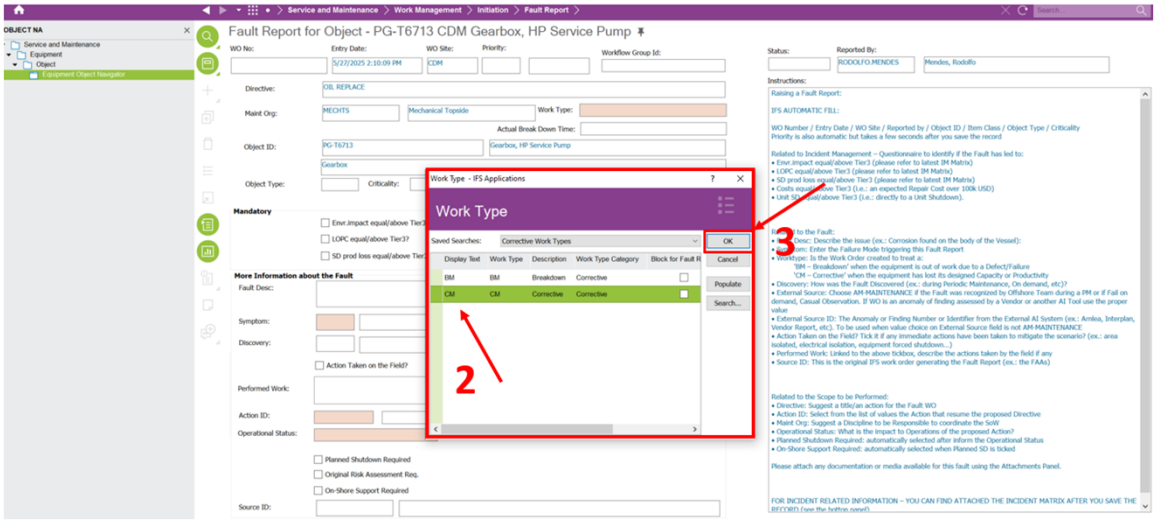


Figure 9 - Confirming the selection of the work type

5.8 EXTERNAL SOURCE

This field refers to the source of information that originated in the request to raise the WO. The **external source** must be selected according to the offline monitoring technique, such as lubricant analysis or vibration, and the alert criticality level described in the CBM report (L – LOW, M – MEDIUM or H – HIGH).

The tables below are all the standardized options:

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External Source	External Source Description	Monitoring Technique	Object Type	Latest Finish (Days)
CBM-VIB/H	CBM Vibration Analysis High	Vibration Analysis	SCE	14
CBM-VIB/H	CBM Vibration Analysis High	Vibration Analysis	SECE	14
CBM-VIB/H	CBM Vibration Analysis High	Vibration Analysis	PCE	21
CBM-VIB/H	CBM Vibration Analysis High	Vibration Analysis	BCE	21
CBM-VIB/H	CBM Vibration Analysis High	Vibration Analysis	ECE	14
CBM-VIB/H	CBM Vibration Analysis High	Vibration Analysis	NCE	30
CBM-VIB/M	CBM Vibration Analysis Medium	Vibration Analysis	SCE	30
CBM-VIB/M	CBM Vibration Analysis Medium	Vibration Analysis	SECE	30
CBM-VIB/M	CBM Vibration Analysis Medium	Vibration Analysis	PCE	45
CBM-VIB/M	CBM Vibration Analysis Medium	Vibration Analysis	BCE	45
CBM-VIB/M	CBM Vibration Analysis Medium	Vibration Analysis	ECE	30
CBM-VIB/M	CBM Vibration Analysis Medium	Vibration Analysis	NCE	60
CBM-VIB/L	CBM Vibration Analysis Low	Vibration Analysis	SCE	60
CBM-VIB/L	CBM Vibration Analysis Low	Vibration Analysis	SECE	60
CBM-VIB/L	CBM Vibration Analysis Low	Vibration Analysis	PCE	75
CBM-VIB/L	CBM Vibration Analysis Low	Vibration Analysis	BCE	75
CBM-VIB/L	CBM Vibration Analysis Low	Vibration Analysis	ECE	60
CBM-VIB/L	CBM Vibration Analysis Low	Vibration Analysis	NCE	90

Table 1 - Vibration Analysis

External Source	External Source Description	Monitoring Technique	Object Type	Latest Finish (Days)
CBM-LUB/H	CBM Lubricant Analysis High	Lubricant Analysis	SCE	14
CBM-LUB/H	CBM Lubricant Analysis High	Lubricant Analysis	SECE	14
CBM-LUB/H	CBM Lubricant Analysis High	Lubricant Analysis	PCE	21

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CBM-LUB/H	CBM Lubricant Analysis High	Lubricant Analysis	BCE	21
CBM-LUB/H	CBM Lubricant Analysis High	Lubricant Analysis	ECE	14
CBM-LUB/H	CBM Lubricant Analysis High	Lubricant Analysis	NCE	30
CBM-LUB/M	CBM Lubricant Analysis Medium	Lubricant Analysis	SCE	30
CBM-LUB/M	CBM Lubricant Analysis Medium	Lubricant Analysis	SECE	30
CBM-LUB/M	CBM Lubricant Analysis Medium	Lubricant Analysis	PCE	45
CBM-LUB/M	CBM Lubricant Analysis Medium	Lubricant Analysis	BCE	45
CBM-LUB/M	CBM Lubricant Analysis Medium	Lubricant Analysis	ECE	30
CBM-LUB/M	CBM Lubricant Analysis Medium	Lubricant Analysis	NCE	60
CBM-LUB/L	CBM Lubricant Analysis Low	Lubricant Analysis	SCE	60
CBM-LUB/L	CBM Lubricant Analysis Low	Lubricant Analysis	SECE	60
CBM-LUB/L	CBM Lubricant Analysis Low	Lubricant Analysis	PCE	75
CBM-LUB/L	CBM Lubricant Analysis Low	Lubricant Analysis	BCE	75
CBM-LUB/L	CBM Lubricant Analysis Low	Lubricant Analysis	ECE	60
CBM-LUB/L	CBM Lubricant Analysis Low	Lubricant Analysis	NCE	90

Table 2 - Lubricant Analysis

CMMS STEP BY STEP

I. Click on the "Values List" available in the CMMS to open the selection window.

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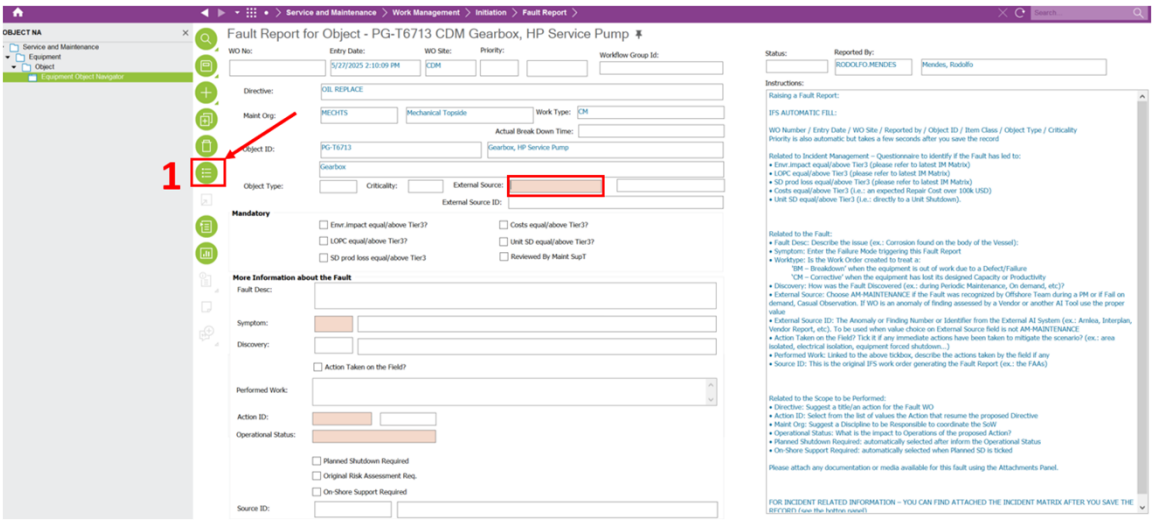


Figure 10 - Field to populate the external source

II. Select the desired option from the selection window.

III. After making your selection, click "OK" to confirm the transaction, as shown below.

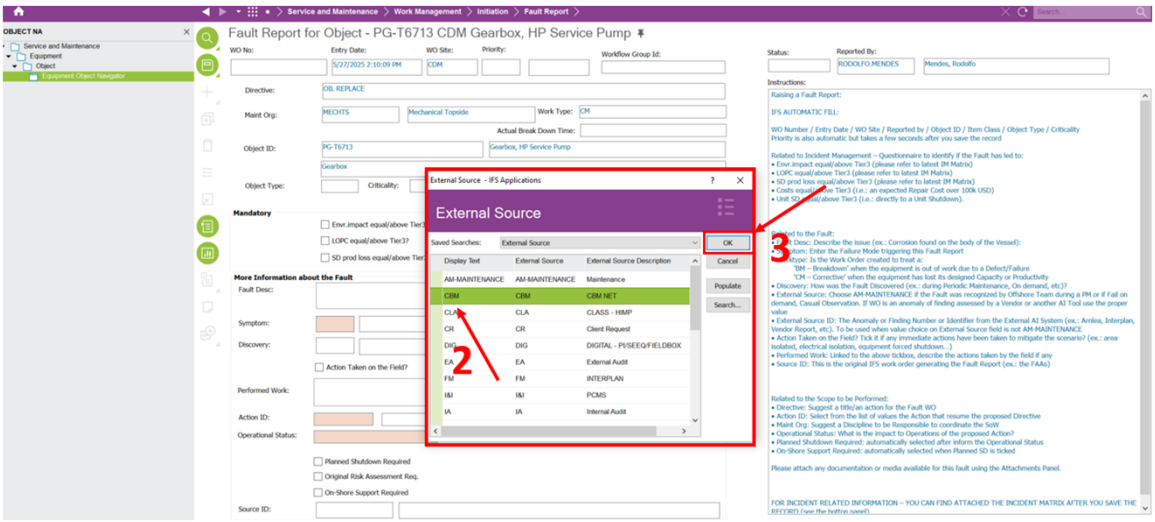


Figure 11 - Confirming the selection of the external source

5.9 EXTERNAL SOURCE ID

This field refers to the code for the information source (External Source). If the source is a CBM report, the External Source ID will always be the action number issued by the platform that manages the predictive report.

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CMMS STEP BY STEP

I. To populate this field, copy the number of External Source ID available in the CBM report, click on the location below, and paste the number in the External Source ID field.

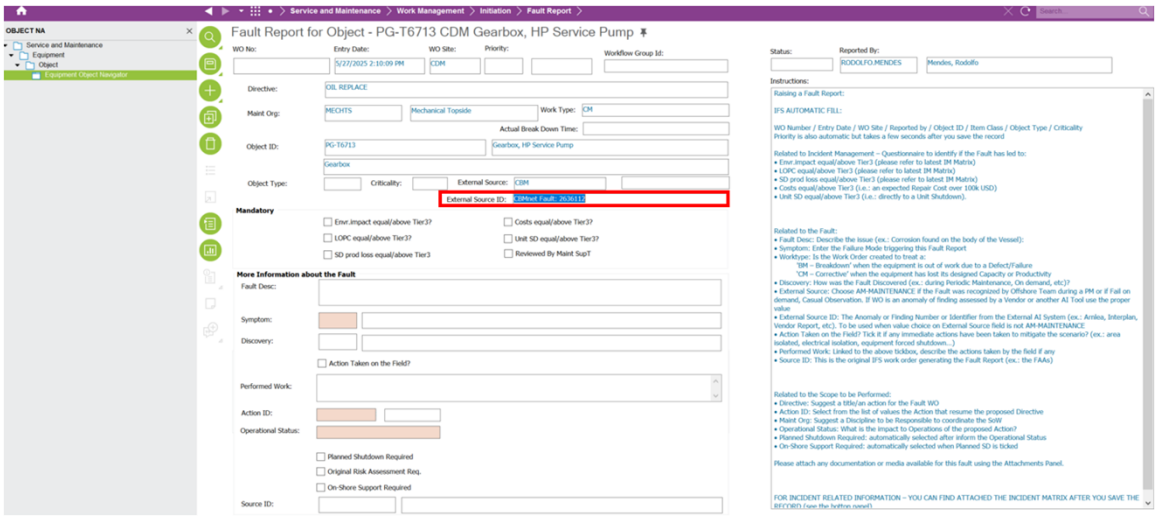


Figure 12 - Field to populate the external source ID

5.10 FALT DESC.

This field refers to the description of the anomaly identified during predictive monitoring. This field should be populated in a structured and organized manner with as much relevant information as possible for the successful completion of the work. The minimum information that should be included in the Fault Description field are:

A. A brief description of the anomaly reported in the CBM reports.

B. Detailed technical recommendation from the engineer responsible for raising the WO in the CMMS.

C. Where the recommendation originates from the lubricant analysis monitoring technique, include the identification number of the sampling point, if applicable.

Additional information can be included to optimize task execution and assist the maintenance team in fully understanding the service. After completing the detailed description of the work, the user should click "OK" to finalize the transaction.

CMMS STEP BY STEP

I. Click on the “Falt Desc.” field to display the edit window [Falt Desc.].

II. Describe the technical recommendations following the guidelines above.

III. Where the recommendation originates from the lubricant analysis monitoring technique, include the identification number of the sampling point, if applicable.

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IV. After that, click "OK" to complete the transaction, as shown below.

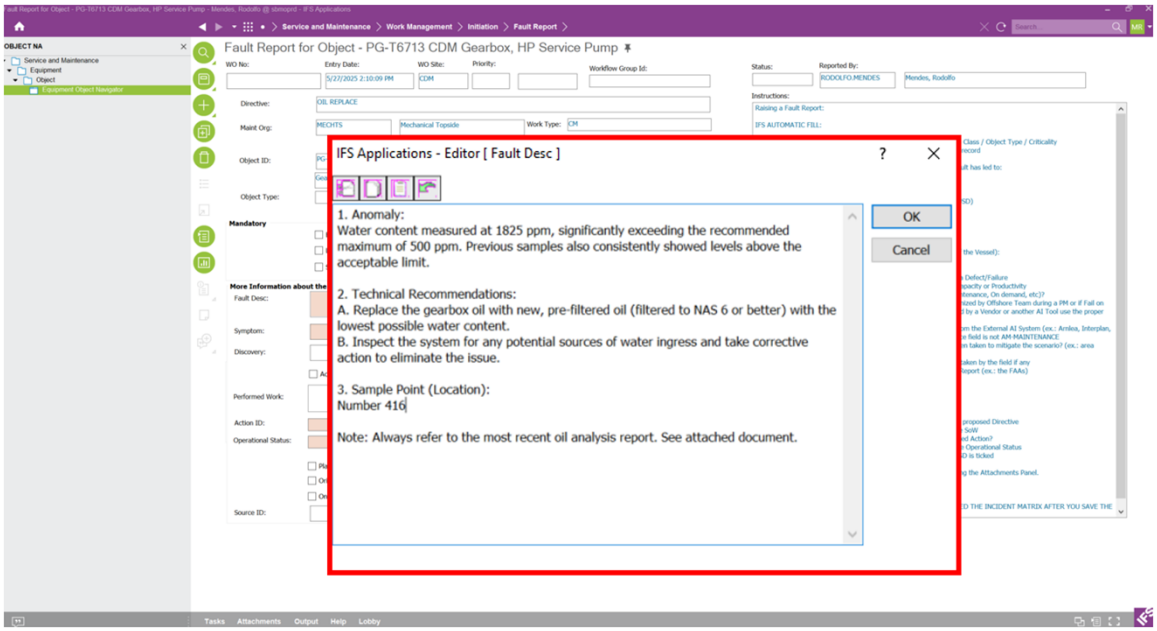


Figure 13 - Field to populate the fault desc.

5.11 SYMPTOM

The symptom code indicates the specific symptom associated with the reported fault. Although various options are available in the CMMS, the user responsible for raising the WO should evaluate the best option considering the symptom identified by predictive monitoring. The options may vary depending on the asset family.

CMMS STEP BY STEP

I. Click on the "Values List" available in the CMMS to open the selection window.

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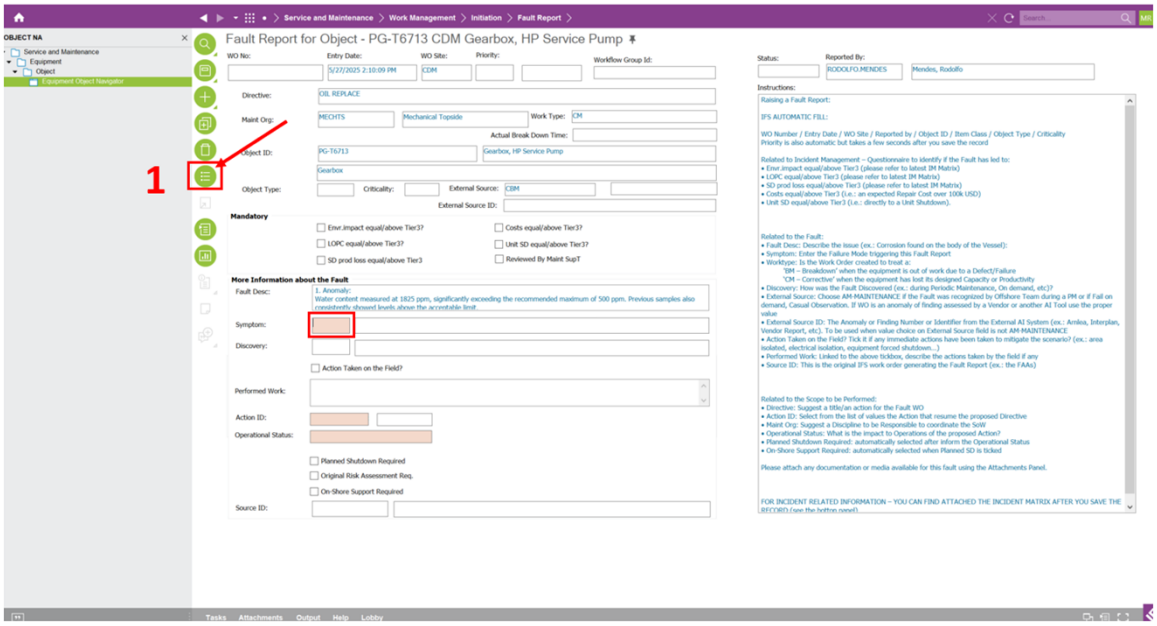


Figure 14 - Field to populate the symptom

II. Select the desired option from the selection window.

III. After making your selection, click "OK" to confirm the transaction.

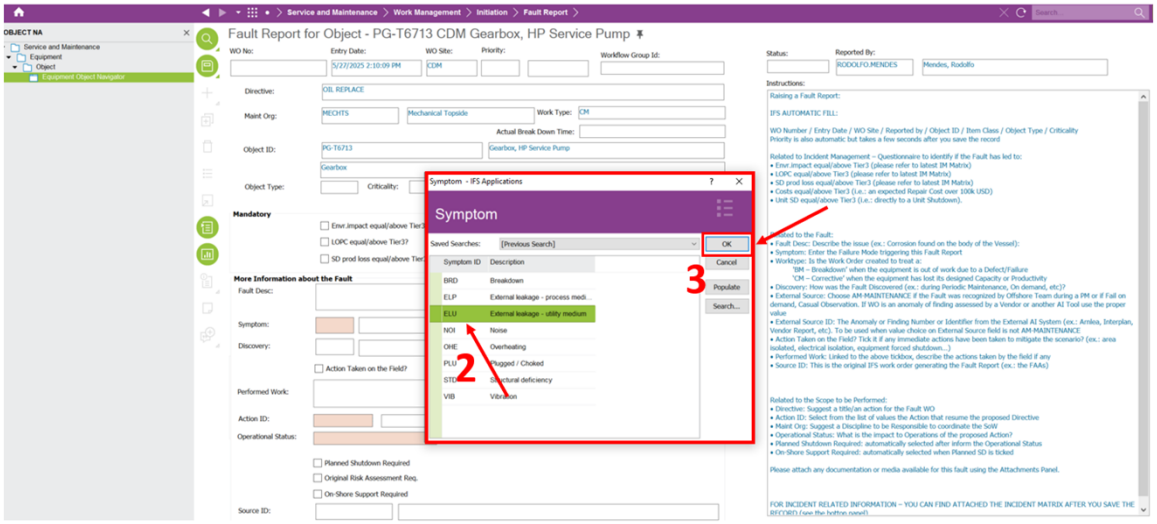


Figure 15 - Confirming the selection of the symptom

5.12 DISCOVERY

This field refers to the method used to identify the anomaly Although various options are available in the CMMS, for raising WO originating from predictive monitoring, option 04 (Periodic condition monitoring) shall ALWAYS be selected.

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CMMS STEP BY STEP

I. Click on the "Values List" available in the CMMS to open the selection window.

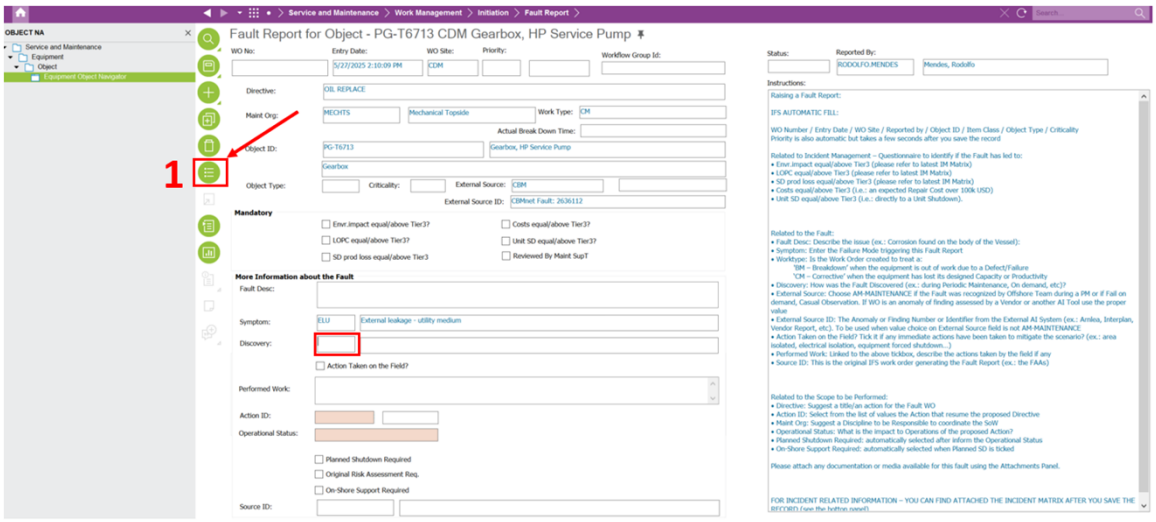


Figure 16 - Field to populate the discovery

II. Select the desired option (Periodic condition monitoring) from the selection window.

III. After making your selection, click "OK" to confirm the transaction.

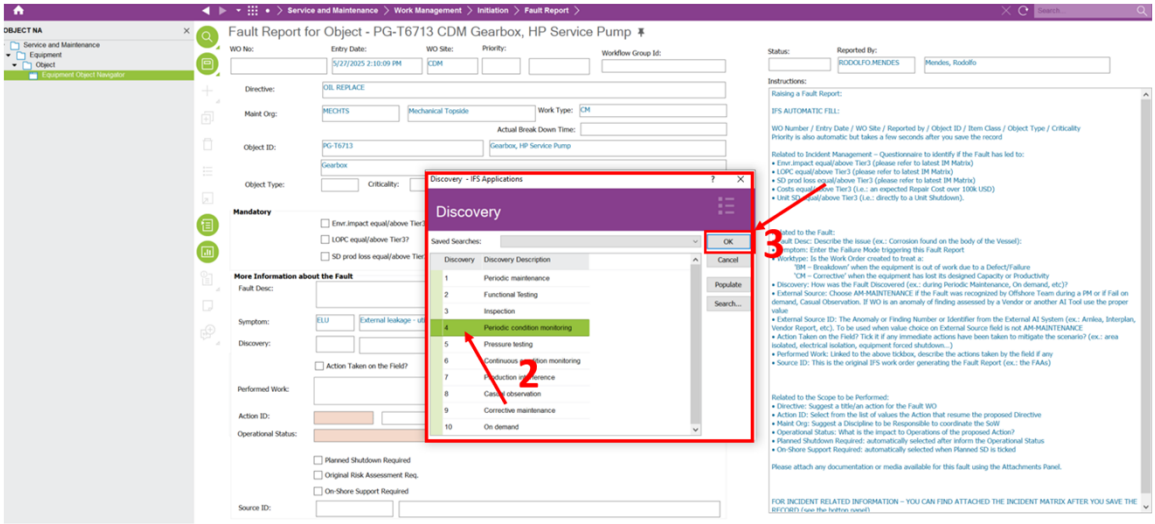


Figure 17 - Confirming the selection of the discovery

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5.13 ACTION ID

This field refers to the type of service to be performed by the maintenance team. The person responsible for raising the WO should evaluate and select the most appropriate option, considering the description of the technical recommendations made by the responsible engineer and the failure modes described in the CBM report.

CMMS STEP BY STEP

I. Click on the "Values List" available in the CMMS to open the selection window.

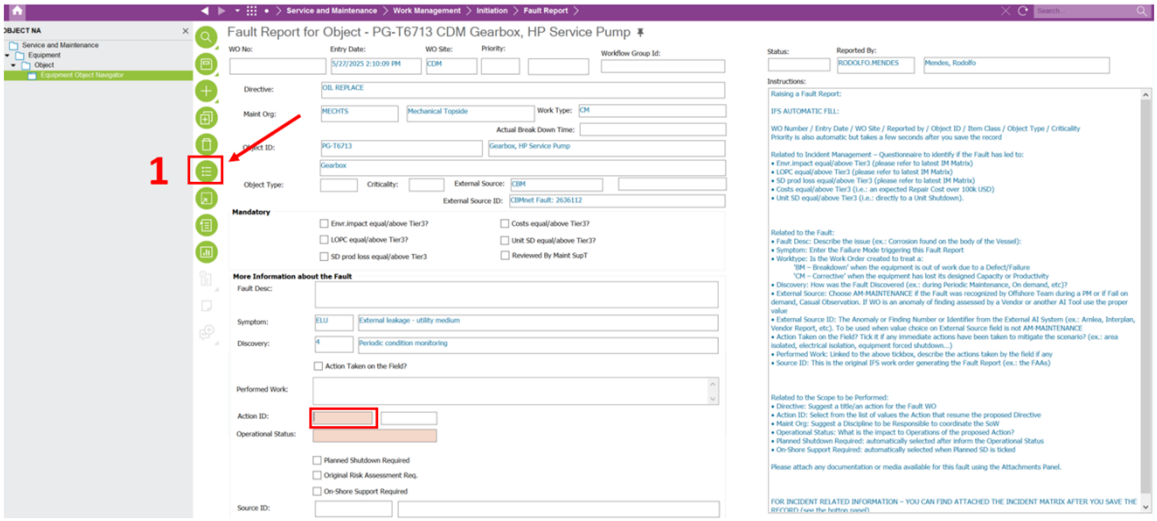
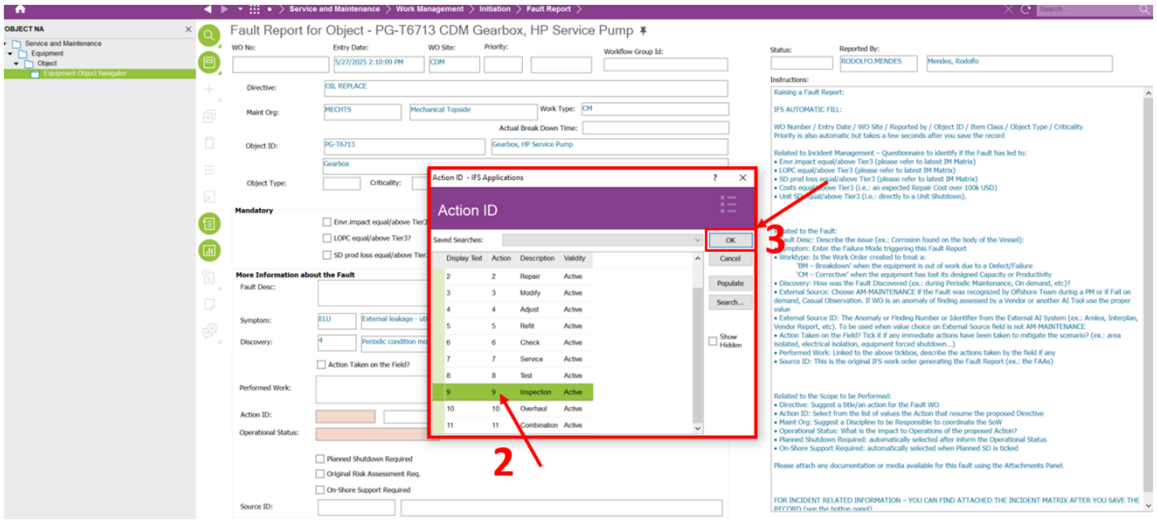


Figure 18 - Field to populate the action ID

II. Select the desired option from the selection window.

III. After making your selection, click "OK" to confirm the transaction, as shown below.



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Figure 19 - Confirming the selection of the action ID

5.14 OPERATIONAL STATUS

This field refers to the type of intervention to be carried out on the equipment. This field will indicate to the maintenance team whether the intervention on the equipment will be intrusive or non-intrusive.

Intrusive:

Intrusive maintenance is maintenance that makes the function of the equipment or system unavailable.

Non-intrusive:

Non-intrusive maintenance refers to maintenance that does not disrupt the operation of the equipment or system.

CMMS STEP BY STEP

I. Click on the "Values List" available in the CMMS to open the selection window.

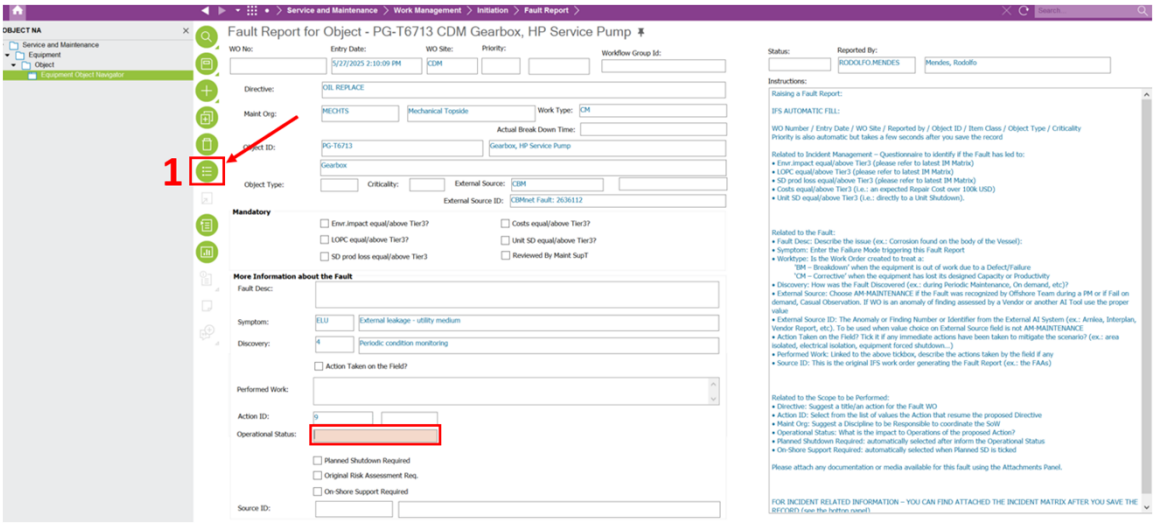


Figure 20 - Field to populate the operational status

II. Select the desired option from the selection window.

III. After making your selection, click "OK" to confirm the transaction.

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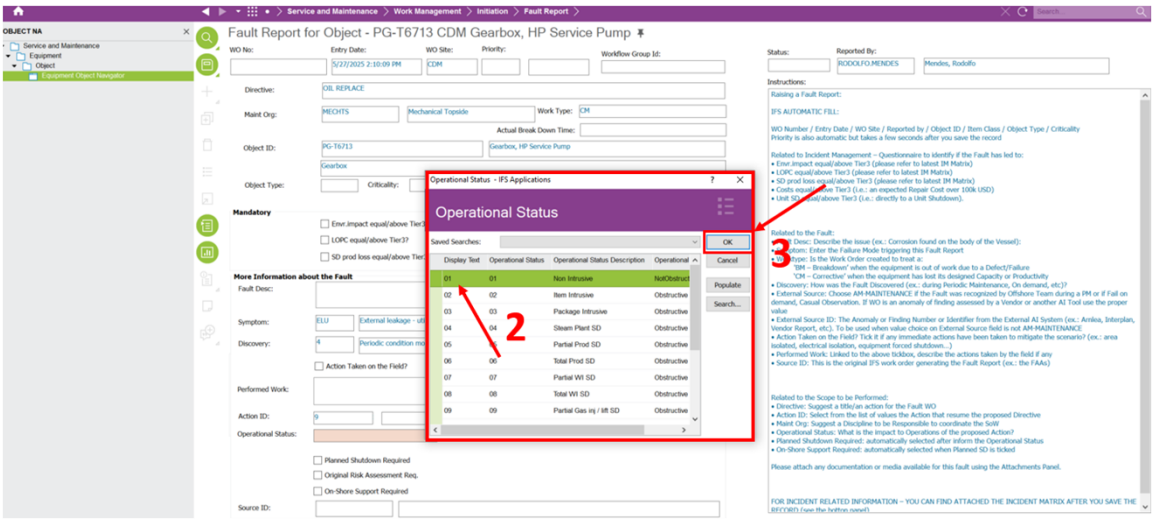


Figure 21 - Confirming the selection of the operational status

5.15 SAVE THE TRANSACTION IN THE CMMS

This This step refers to “saving the operation”. After populating all the required fields, the operation needs to be saved in the CMMS to create a WO number.

CMMS STEP BY STEP

I. Click on the "Save" option

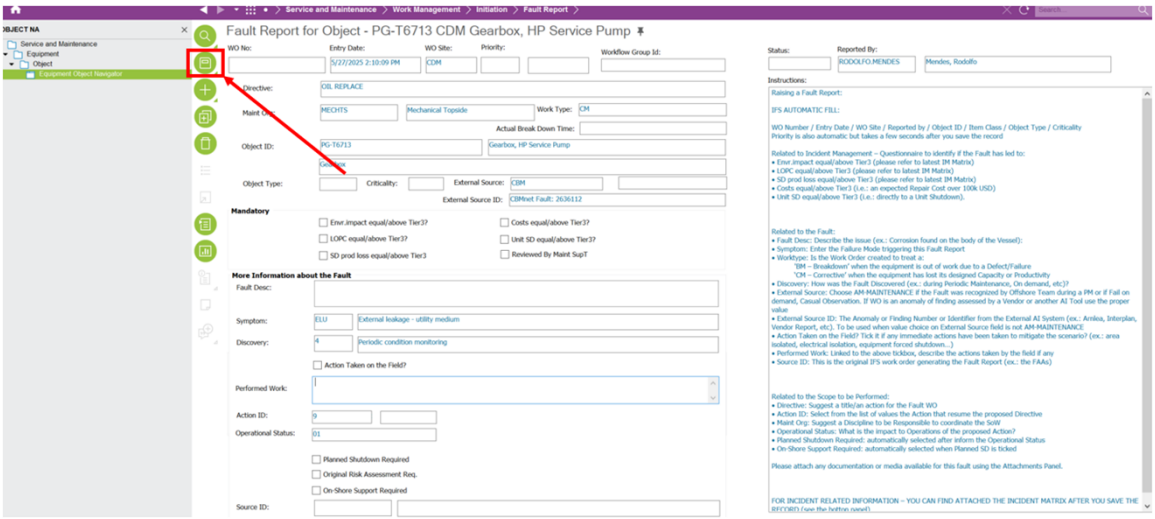


Figure 22 - Transaction saving process

II. Verify that the WO number has been generated, as shown below.

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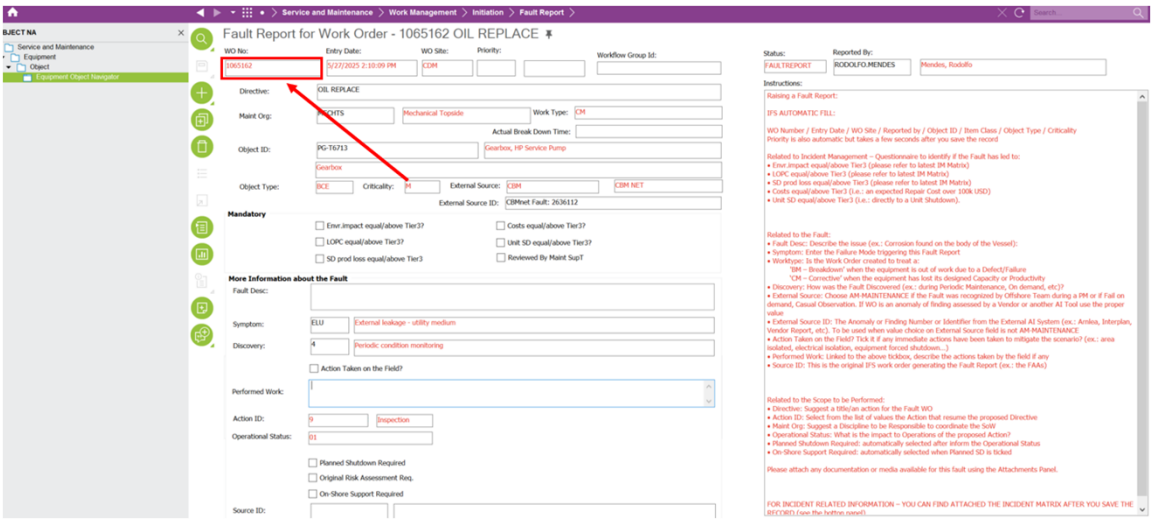


Figure 23 - Verification of the WO number

5.16 ATTACH THE CBM REPORT IN THE CMMS

This step refers to the uploading of CBM report. After saving the transaction and generating the Work Order, the CBM report must be attached to the CMMS.

CMMS STEP BY STEP

I. After confirming the creation of the Work Order, click on "Attachments," then click on "Documents," and finally click on "New."

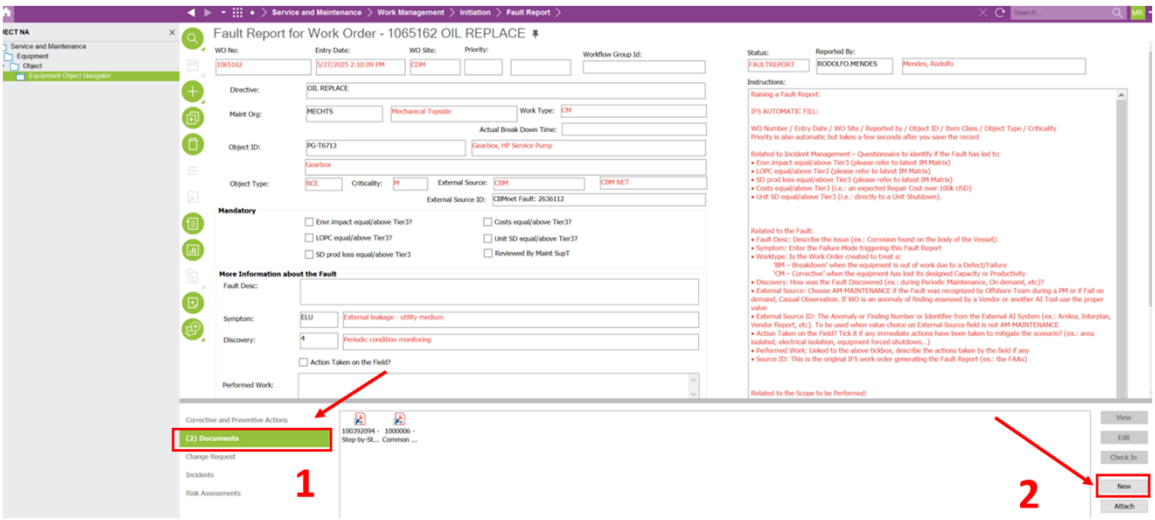


Figure 24 - Attaching the CBM report - Step 01

II. Next, click on the "Document Class" field to select the type of document that will be attached to the Work Order.

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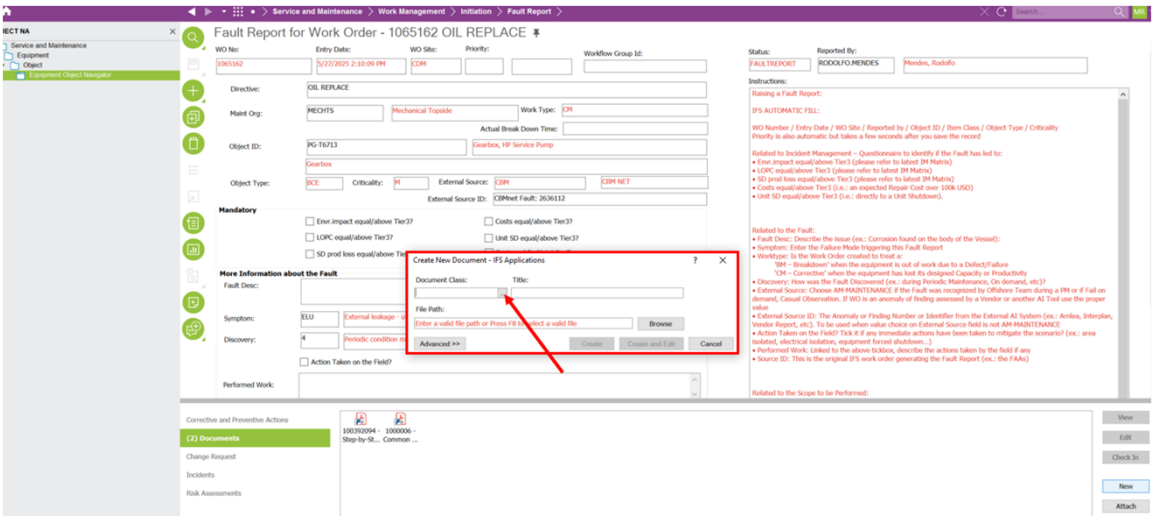


Figure 25 - Attaching the CBM report - Step 02

III. To attach CBM reports, always select "Document Class" LS-042 and select “OK”.

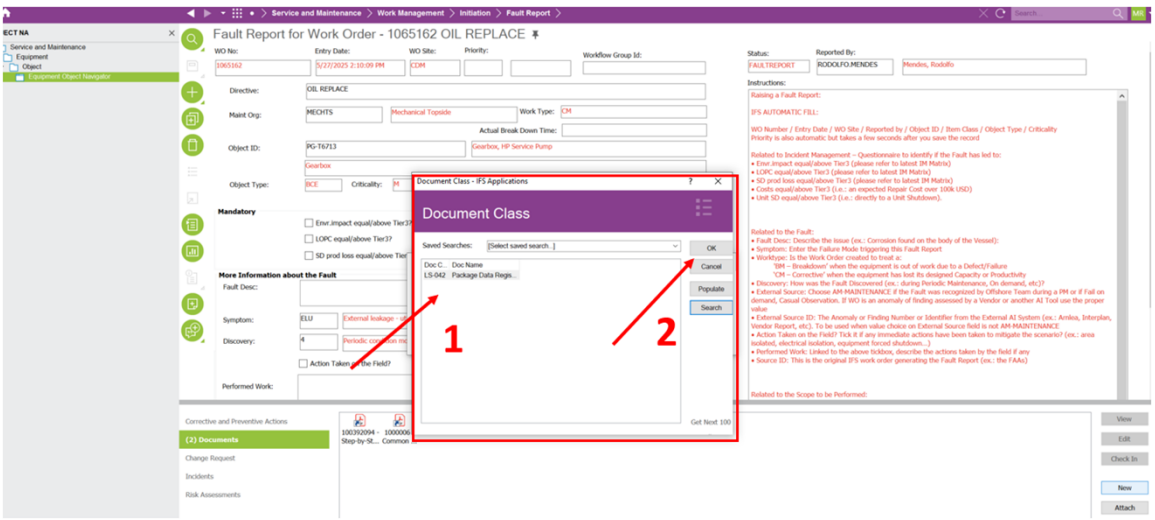


Figure 26 - Attaching the CBM report - Step 03

IV. Next, click on "Browse" and select the file (CBM Report) from your computer and click on “Create”.

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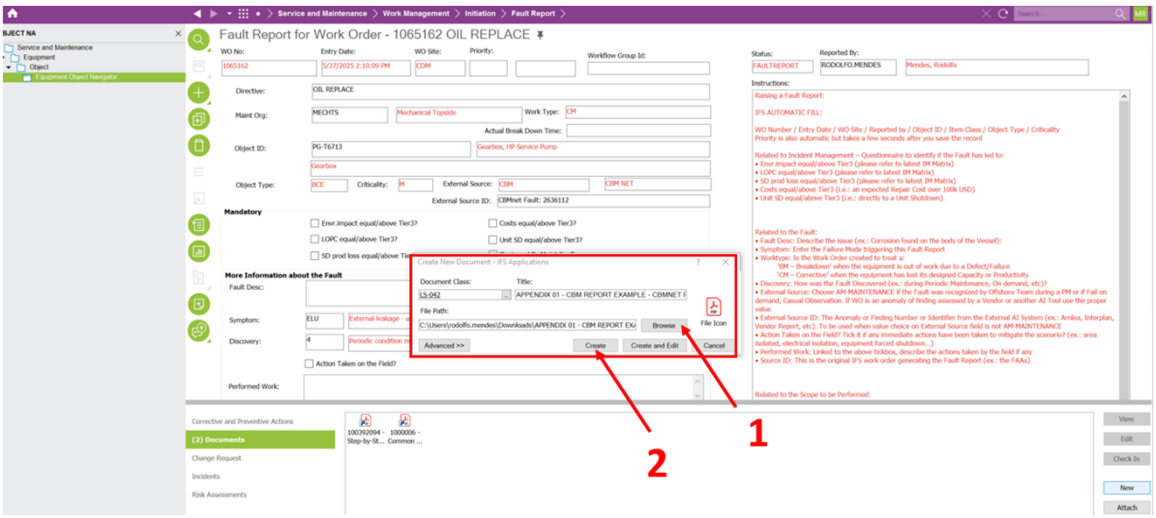


Figure 27 - Attaching the CBM report - Step 04

V. Finally, make sure that the report has been attached correctly.

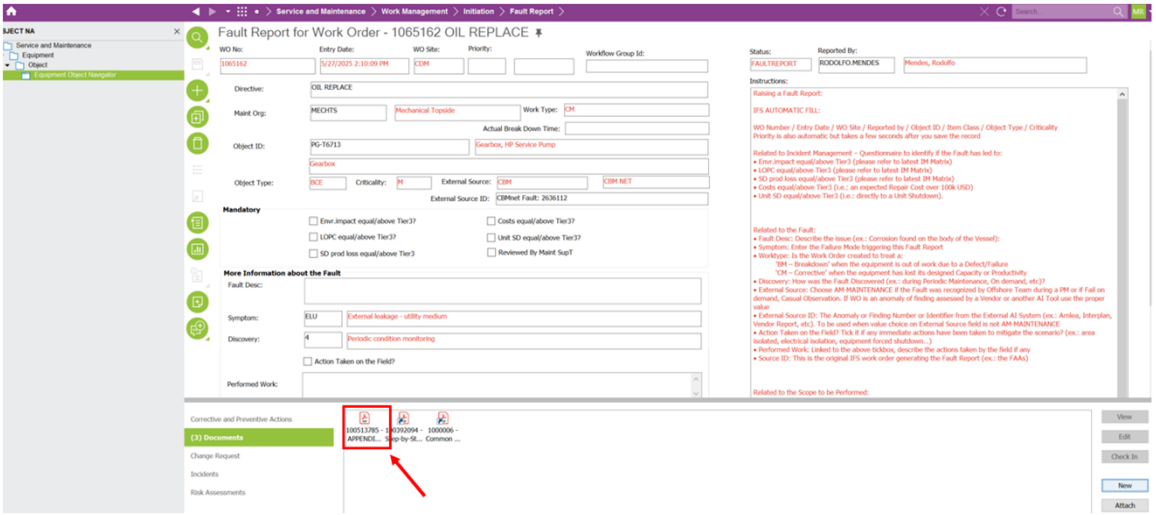


Figure 28 - **Confirming that the report has been attached**

NOTE: An example of the CBM report can be found in the APPENDICES.

6. WORK ORDER CLOSURE STEPS

After the WO is generated in the CMMS, the supervisor verifies the accuracy of the information, checks for any duplicates, and ensures the correct assignment of discipline. If the information is accurate, the supervisor changes the status of the WO to **"Observed"**. If the WO is found to be inadequate or incorrectly created, the supervisor

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shall contact who created the WO to get the proper clarification required.

Once validated, the supervisor is responsible for entering the total duration of the task in the "Duration" field and providing the detailed scope of work in the "Work Description" field. If necessary, a Risk Assessment must also be created.

Next, the Maintenance Superintendent reviews the information and updates the WO status to “**Under Preparation**”. In this status, all WOs are considered confirmed, and fields become locked for editing. The supervisor proceeds to request the required resources: for materials, a material request is issued, and if the item is not in stock, a purchase requisition must be created. For services, a purchase requisition is always required.

Once all resource requests are **completed** (materials, services, and manpower via the Resources tab), the status is changed to “Prepared”. If materials or services are already available, the supervisor may directly update the status to “**Released**”.

Subsequently, the “**Work Started**” status is automatically triggered when the actual start date is entered, indicating the beginning of the task. Upon completion, the supervisor updates the status to “**Work Done**”, records the work details in the "**Performed Work**" field, logs the man-hours used, and attaches the relevant reports.

If there's an open PO, it’s necessary to set as technically completed to finalize the WO. If the PO it’s already close or the WO don’t have PO, the supervisor changes to “**Reported**”.

Finally, the Maintenance Superintendent or Onshore Maintenance Engineer must evaluate whether the service was executed correctly, according to all technical recommendations outlined in the “Fault Description”, and then update the status to “**Finished**”, thus closing the WO lifecycle in the CMMS.

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7. RACI MATRIX

#	Activity	Reliability Engineer / Vibration Specialist	Offshore Maintenance Engineer	Maintenance Superintendent	Maintenance Supervisor	Predictive Company (3rd)
1	Open a predictive maintenance work order in CMMS	R	I	I	I	C
2	Verify the accuracy of the WO	C	I	A	R	
3	Evaluate if the service was properly executed	I	A	R	C	

8. SUPPORTING DOCUMENTS

There is no supporting document for this process.

9. APPENDICES

APPENDIX A. CBM REPORT EXAMPLE - CBMNET REPORT
APPENDIX B. CBM REPORT EXAMPLE - LABIT REPORT
APPENDIX C. CBM REPORT EXAMPLE - MOBIL SERV REPORT